

ILLINOIS BASKETBALL

Lineup Synergy Optimizer

Analytics Internship Submission · Ajay Suresh · June 2026

1. What Problem Are We Solving?

College basketball coaches make hundreds of lineup decisions across a season - who starts, who closes games, who pairs with whom in crunch time. These decisions are typically made on intuition, film, and feel. The challenge is that even experienced coaches can't evaluate all possible lineup combinations simultaneously, especially when trying to account for role synergies, usage distribution, and floor spacing across a full roster. This tool aims to solve these complexities.

The Core Question

Given Illinois's current roster, which 5-man lineups are projected to maximize adjusted efficiency margin and why do certain combinations work better than others?

The tool also addresses a second, equally important problem: the transfer portal. In the modern era, roster construction happens year round. Coaches need to quickly assess whether a portal target fits their system before making a scholarship offer. The Portal Fit Analyzer gives coaching staff a data-driven first look.

2. Data Sources & Methodology

Data Source

All data was sourced from BartTorvik.com, the leading public repository for college basketball advanced analytics. BartTorvik provides player-level efficiency metrics, role classifications, and team-level adjusted efficiency ratings for all D-I programs.

We collected five seasons of data (2022–2026), yielding approximately 1,800 team-seasons across all D-I programs. Each team-season includes:

- **Player-level stats:** ORtg, USG%, eFG%, 3PT attempt rate, AST%, TOV%, ORB%, DRB%, BLK%, STL%, and BartTorvik role classification (Pure PG, Combo G, Wing F, Stretch 4, PF/C, C, etc.)
- **Team-level targets:** Adjusted Offensive Efficiency (adjOE), Adjusted Defensive Efficiency (adjDE), and Adjusted Efficiency Margin (adjEM = adjOE – adjDE)

Feature Engineering

For each team-season, we constructed two categories of features:

Individual Quality Features - Minute-weighted averages of player efficiency metrics across the rotation (MIN% ≥ 15%). These capture the talent level of the roster.

Roster Construction & Synergy Features - Inspired by the Sloan Sports Analytics Conference paper “Lineup Optimization via Deep Reinforcement Learning,” which demonstrates that lineup efficiency is not simply the sum of individual talents but includes pairwise interaction effects. Our synergy features include:

- **Usage variance:** measures ball-sharing vs. ball-dominance. The model learned a negative coefficient here, teams where usage is concentrated in one or two players are less efficient on average.
- **Frontcourt 3PT%:** quantifies stretch big value, does the frontcourt space the floor?
- **Guard playmaking depth:** minute-weighted AST% among guards specifically.
- **Role entropy:** Shannon entropy of the role distribution, higher entropy = more versatile roster.
- **Role composition counts:** number of Pure PGs, Stretch 4s, PF/Cs, and Centers on the floor.

3. How We Built the Solution

Model Architecture

We trained four models to predict team adjEM from roster construction features, using temporal cross-validation (TimeSeriesSplit, n=4) to prevent lookahead bias, models are always trained on past seasons and validated on future ones.

Model	CV R ² (Mean ± Std)	CV MAE (adjEM pts)
Bayesian Ridge (+ Lasso feature selection)	0.645 ± 0.017	6.09 ± 0.57
Elastic Net (CV-tuned)	0.647 ± 0.019	6.13 ± 0.56
XGBoost	0.589 ± 0.032	6.65 ± 0.66
Random Forest	0.605 ± 0.027	6.37 ± 0.54

The Bayesian Ridge model was selected as the primary model for two reasons: it matches Elastic Net performance while providing posterior uncertainty estimates (credible intervals) on lineup projections and its coefficients are directly interpretable as feature importance without post-hoc analysis. Furthermore, it uses less features compared to Elastic Net, therefore providing a gain in terms of model interpretability. Please see Exhibit 2 for further information regarding the chosen model’s predictive performance.

Lasso Feature Selection

Multicollinearity between offensive metrics (ORtg, eFG%, TS%) was addressed with a two-stage pipeline: LassoCV first selects the non-redundant feature subset via L1 regularization, then Bayesian Ridge estimates coefficients on the selected features. Exhibit 1 captures the magnitude of the coefficients for the final set of features within the Bayesian Ridge model.

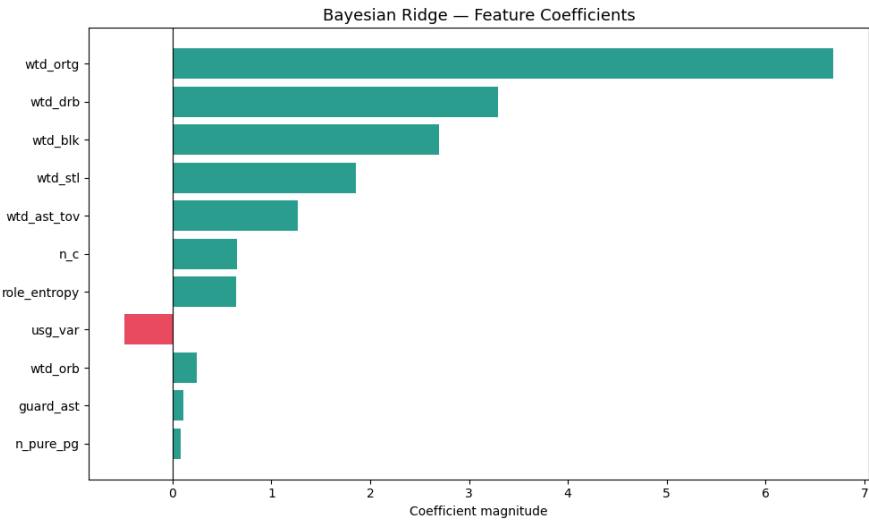


Exhibit 1: Feature importance plot of Bayesian Ridge Model

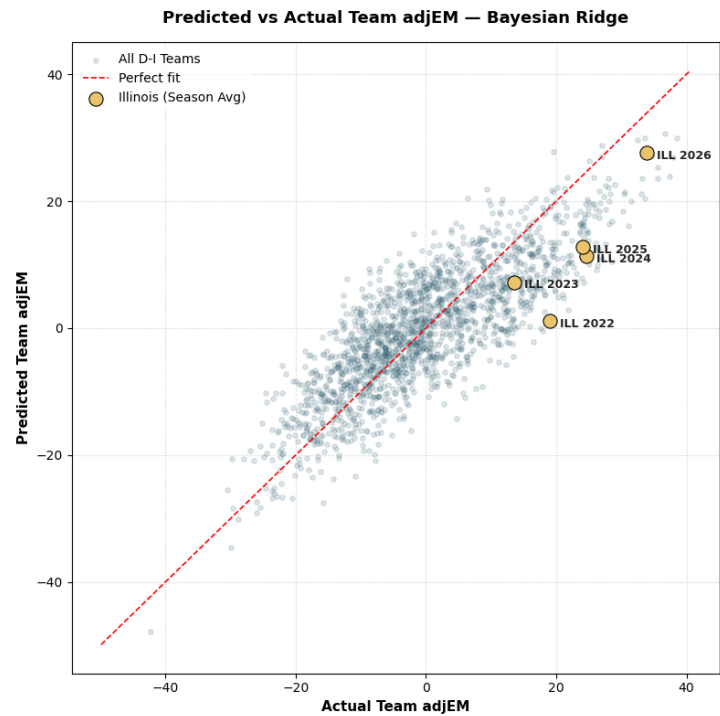


Exhibit 2: Predicted adjEM vs Actual adjEM

Lineup Projection

Given any 5 man lineup, the app constructs the same minute-weighted feature vector and passes it through the trained pipeline to produce a projected adjEM with a 95% credible interval. All $C(n, 5)$ possible lineups for a given roster are enumerated and ranked.

Limitations

Lineup projections are relative comparisons. The model was trained on full-rotation team features; applying it to 5-player subsets introduces a systematic upward bias in absolute adjEM values. Rankings across lineups are reliable, absolute numbers should be treated directionally. Additionally, the model correctly learns that high usage concentration is a net negative at the population level but may understate elite high-usage players (e.g. AJ Dybantsa, Keaton Wagler, Darius Acuff Jr., etc.) whose individual talent exceeds what roster construction metrics can capture.

Technology Stack

- Python: pandas, numpy, scikit-learn (BayesianRidge, LassoCV, SelectFromModel, TimeSeriesSplit), XGBoost
- Data product: Streamlit web application with Plotly interactive charts
- Modeling: Jupyter Notebook (lineup_model.ipynb)
- Data source: BartTorvik.com player and team data, 2022–2026

4. Why This Is Useful

Tab 1: Lineup Rankings

Every possible 5-man lineup from the current roster, ranked by projected adjEM with 95% credible intervals. A coach can instantly see which lineups the model projects as most efficient, and how much uncertainty surrounds each projection.



Tab 2: Lineup Simulator

A coach selects any 5 players and sees an instant projection alongside contextual flags: usage congestion warnings when two high-USG players share the floor, spacing assessments based on frontcourt 3PT tendencies, and a gauge showing where the lineup sits relative to all D-I teams.

BUILD ANY 5-MAN LINEUP

Select exactly 5 players to see the projected adjEM and credible interval. Test lineups that have never played together.

Select 5 players

Kylan Boswell

Jake Davis

Zvonimir Ivisic

Keaton Wagler

David Mirkovic

PROJECTED ADJEM

+29.0

95% CI: [15.1, 42.9]

$\sigma = 7.08$

LINEUP RANK

#26

of 56

Among all possible 5-man combos

PLAYER BREAKDOWN

Player	Role	MIN%	OBPM	DBPM	ORTg	3P%	USG%	eFG%
Kylan Boswell	Combo G	60.1	5.93	1.39	126.3	30.7	20.6	50.7
Jake Davis	Wing G	48.3	6.38	1.95	146.6	40.6	10.9	61.2
Zvonimir Ivisic	PF/C	42.4	2.93	7.51	127.2	28	16.2	59.1
Keaton Wagler	Combo G	83.7	8.84	2.32	130.2	39.7	25.7	54.1
David Mirkovic	Stretch 4	72.7	5.86	2.99	122.7	37.5	23.3	56.2

SYNERGY NOTES

✓

Good spacing: Frontcourt avg 3PT% is 33%.

💡

See the Synergy Explorer tab for data-driven role pair analysis for this roster.

Tab 3: Synergy Explorer

A heatmap of role-pair synergy delta used to determine which role player combinations produce lineups that outperform expectations. A player impact table shows the average adjEM of lineups with vs. without each player, surfacing which players are most valuable to roster construction.

ROLE SYNERGY HEATMAP

For each role pair, this shows how the projected adjEM changes when both roles are present in a lineup vs. when only one is. Positive = complementary. Negative = redundant.

Wing G

Wing F

Stretch 4

PF/C

Combo G

-6.3

-6.3

-1.5

1.5

0.0

-0.8

-0.9

10.0

0.0

1.5

-2.2

-2.3

0.0

10.0

-1.5

-5.6

0.0

-2.3

-0.9

-6.3

0.0

-5.6

-2.2

-0.8

-6.3

Combo G

PF/C

Stretch 4

Wing F

Wing G

adjEM delta

5

0

-5

Note: synergy deltas are computed from projected adjEM across the possible lineups. Small sample sizes per role pair mean these estimates carry uncertainty — treat as directional, not definitive.

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Tab 4: Portal Fit Analyzer

A coach searches any player in the 2026 dataset by name, sees their efficiency profile, and gets an instant fit assessment: role redundancy on current roster, usage compatibility, and spacing value. They then select 4 current players to build a hypothetical lineup with the portal target and see the projection ranked against the team's existing combinations.

TRANSFER PORTAL FIT ANALYZER

Assess how a transfer portal target fits your system. Select a portal player to see their profile, role fit, and how they'd project in hypothetical lineups with your current roster.

Search portal player by name

Stefan vaa

Select portal target

Stefan Vaaks — Providence (Combo G, ORtg 115.4, BPM 2.8)

PLAYER PROFILE

Player	Current Team	Role	MIN%	BPM	OBPM	DBPM	ORtg	3P%	USG%	eFG%	BLK%	DRB%	AST%
Stefan Vaaks	Providence	Combo G	72.1	2.81	4.77	-1.96	115.4	35	23.4	52.2	0.8	6.7	18.1

ROLE FIT ASSESSMENT

SAME ROLE ON ROSTER

2

Combo G already rostered

USAGE COMPATIBILITY

⚠ High

Portal USG% 23 · 3 high-usage players on roster

SPACING VALUE

—

Stretch big who shoots threes

5. Links

Live Application:

Deployed on Streamlit Community Cloud — <https://cbblineupoptimizer.streamlit.app>

GitHub Repository: https://github.com/Linkforce1/cbb_lineup_optimizer

The repository includes the full modeling notebook (notebooks/lineup_model.ipynb), Streamlit application (app/streamlit_app.py), and all data files (data/).